Gravitational Fields Summary

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1 Formulae

1.1 Standard

1. Gravitational Field Strength

$$g = \frac{F}{m} \tag{1}$$

or more useful

$$g = -\frac{GM}{r^2} \tag{2}$$

2. Newton's Law of Gravitation

$$F_g = \frac{GMm}{r^2} \tag{3}$$

3. Gravitational Potential

$$V_g = -\frac{GM}{r} \tag{4}$$

4. Change in gravitational potential

$$\Delta V_g = \frac{\Delta W}{m} \tag{5}$$

5. Potential Gradient

$$g = -\frac{\Delta V_g}{\Delta r} \tag{6}$$

1.2 Derived

1. Orbital Time Period

   On a planet, $\sum F = F_g$. Hence:

$$\begin{aligned} F_g &= F_c \\ \frac{GMm}{r^2} &= m\omega^2 r \\ \omega^2 &= \frac{GM}{r^3} \\ \frac{4\pi^2}{T^2} &= \frac{GM}{r^3} \\ T^2 &= \frac{4\pi^2}{GM} r^3 \end{aligned} \tag{7}$$

   Kepler's 3rd Law: $T^2 \propto r^3$

$$T_{ORB} = \sqrt{\frac{4\pi^2 r^3}{GM}}$$

2. Orbital Velocity ($\neq$ Escape Velocity)

   Using the fact that $m\omega^2 r = \frac{mv^2}{r}$

$$\begin{aligned} F_g &= F_c \\ \frac{GMm}{r^2} &= \frac{mv^2}{r} \\ \frac{GM}{r} &= v^2 \\ \therefore v_{ORB} &= \sqrt{\frac{GM}{r}} \end{aligned} \tag{8}$$

3. Escape Velocity.

   When work is done on a mass to move it to infinity, the mass' Gain in E_g = Loss in E_k. Hence:

$$\begin{aligned} \Delta W &= \Delta E_k \\ mV_g &= \frac{1}{2}mv^2 \\ \frac{GM}{r} &= \frac{1}{2}v^2 \\ \therefore v_{ESC} &= \sqrt{\frac{2GM}{r}} \\ &= \sqrt{2}v_{ORB} \end{aligned} \tag{9}$$

4. Total Energy Change

$$\begin{aligned} \Delta E_T &= \Delta E_g + \Delta E_k \\ &= mV_g + \frac{1}{2}mv^2 \\ &= -\frac{GMm}{r} + \frac{GMm}{2r} \\ &= -\frac{GMm}{2r} \\ &= -E_k \end{aligned} \tag{10}$$

2 Nuance(s)

Be careful with Δ's.

Eg: Calculating the new radius for a change in gravitational potential

$$\begin{aligned} \Delta V_g &= -\frac{GM}{\Delta r} \\ V_2 - V_1 &\neq -\frac{GM}{r_2 - r_1} \\ &= -GM(\frac{1}{r_2} - \frac{1}{r_1}) \end{aligned} \tag{11}$$

Remember r = radius of planet + height above surface